

Lab experiment-4

4. Crypt Arithmetic Problem

Aim

To solve the crypt arithmetic puzzle, $\text{SEND} + \text{MORE} = \text{MONEY}$.

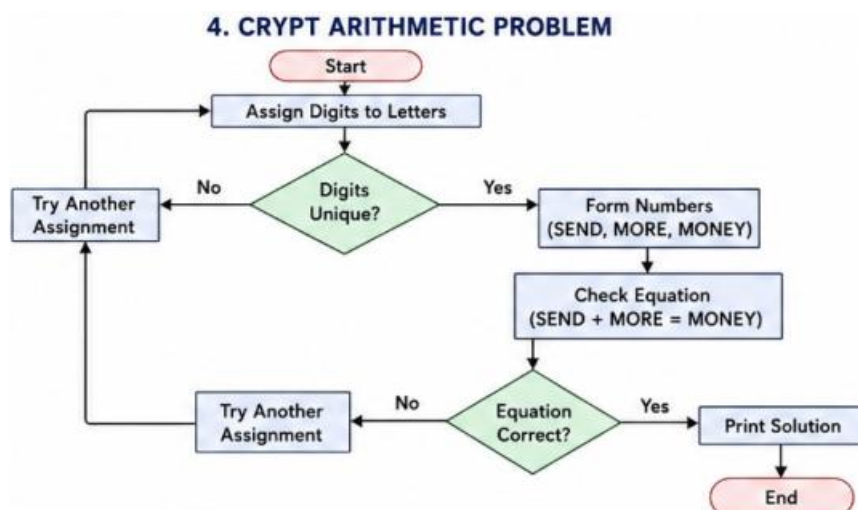
Objective

To assign unique digits to letters so that the arithmetic equation is satisfied.

Algorithm

1. Assign unique digits to each letter.
2. Ensure leading letters are not zero.
3. Form the numbers SEND, MORE, and MONEY.
4. Check whether $\text{SEND} + \text{MORE} = \text{MONEY}$.
5. If true, print the solution; otherwise, try another assignment.

Flowchart



Code

```
from itertools import permutations

letters = "SENDMORY"

for p in permutations("0123456789", 8):

    d = dict(zip(letters, p))

    if d['S'] == '0' or d['M'] == '0':

        continue

    send = int(d['S']+d['E']+d['N']+d['D'])

    more = int(d['M']+d['O']+d['R']+d['E'])

    money = int(d['M']+d['O']+d['N']+d['E']+d['Y'])

    if send + more == money:

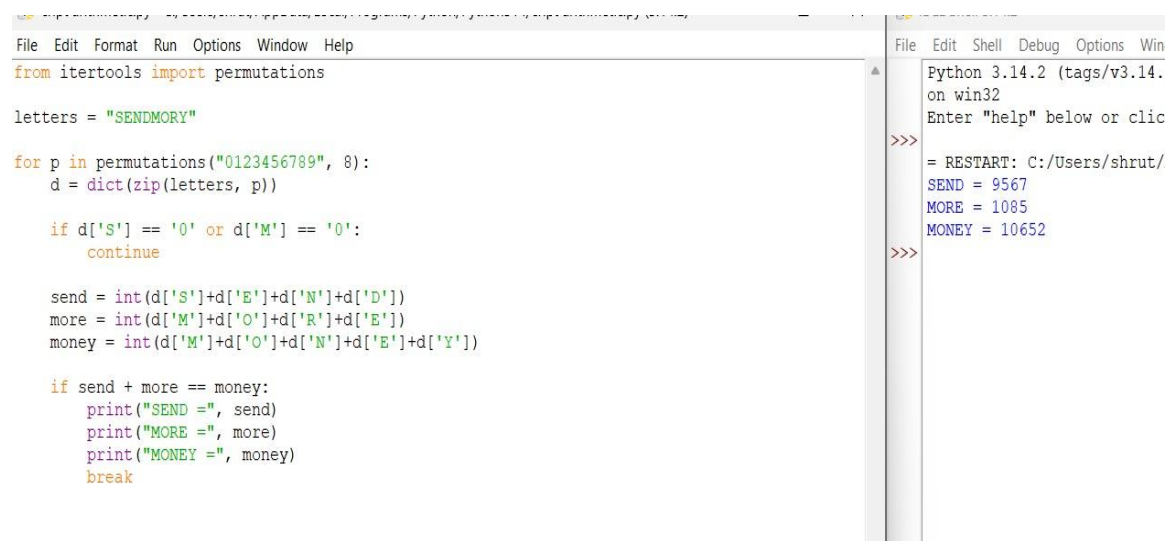
        print("SEND =", send)

        print("MORE =", more)

        print("MONEY =", money)

        break
```

Output

A screenshot of a Python IDE with two panes. The left pane shows the code from the previous block. The right pane shows the output of the program. The output starts with a prompt to enter 'help', followed by a restart message, and then the results: SEND = 9567, MORE = 1085, and MONEY = 10652.

```
File Edit Format Run Options Window Help
from itertools import permutations

letters = "SENDMORY"

for p in permutations("0123456789", 8):
    d = dict(zip(letters, p))

    if d['S'] == '0' or d['M'] == '0':
        continue

    send = int(d['S']+d['E']+d['N']+d['D'])
    more = int(d['M']+d['O']+d['R']+d['E'])
    money = int(d['M']+d['O']+d['N']+d['E']+d['Y'])

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

```
Python 3.14.2 (tags/v3.14.
on win32
Enter "help" below or clic
>>>
= RESTART: C:/Users/shrut/
SEND = 9567
MORE = 1085
MONEY = 10652
>>>
```

Result

The program successfully finds the correct digit assignment.

Conclusion

The Crypt Arithmetic problem is solved by assigning unique digits that satisfy the given equation.

These sections are concise and suitable for writing in an AI laboratory record.